At emission, the satellite distances to both stations are equal:

$$L = c\tau \ . \qquad (2)$$

When satellite P is in the triangle plane, its altitude is maximal, H. Let θ be the latitude of D_1 , D_2. The cosine rule gives

$$L^2 = R^2 + (H+R)^2 - 2R(H+R)\cos\theta \ . \qquad (3)$$

Equations (2), (3) give

$$H = \sqrt{(c\tau)^2 - (R\sin\theta)^2} - R(1-\cos\theta) \ . \qquad (4)$$

Equation (4) also follows directly from the diagram.

(ii) A small latitude error $\triangle\theta$ gives, by (3), an error $\triangle H$ in H. Then we have

$$L^2 = R^2 + (H+\triangle H+R)^2 - 2R(H+\triangle H+R)\cos(\theta+\triangle\theta) \ . \qquad (5)$$

Expand (5). Small θ gives small $\triangle H$. Drop higher orders and subtract (3), giving

$$\triangle H = -\frac{R(R+H)\sin\theta\triangle\theta}{H+(1-\cos\theta)R} \ , \qquad (6)$$

Here H is given by (4).

(iii) An error $\triangle\tau$ in time τ gives an error in L:

$$\triangle L = c\triangle\tau \ . \qquad (7)$$

By (3), this gives an error $\triangle H$ in H. Then

$$(L+\triangle L)^2 = R^2 + (H+\triangle H+R)^2 - 2R(H+\triangle H+R)\cos\theta. \qquad (8)$$

From (7), (8) and 3), neglect higher orders to obtain

$$\triangle H = \frac{c^2\tau\triangle\tau}{H+R(1-\cos\theta)} \ , \qquad (9)$$

Here H is given by (4).

2. (i) From 4), $H = 2.8\times10^{4}$ km. (ii) From 6), $\triangle H = \mp 25$m. (iii) Substituting into 9) gives $H = \pm 3$m .

3. In a chosen frame, let position be x , y , z and time t. Satellite i has coordinates x_i , y_i , z_i at time t_i. Its signal travels at light speed, so

$$(x-x_i)^2 + (y-y_i)^2 + (z-z_i)^2 = c^2(t-t_i)^2 \qquad (i = 1 \ , \ 2 \ , \ 3 \ , \ 4), \qquad (10)$$

(1) has four unknowns t , x , y , z, requiring four independent equations. Receive four or more satellites at once. Position and time follow from (10), i